Sex Roles
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ORIGINAL ARTICLE

Are Gender Differences in Vocational Interests Universal?: Moderating Effects of Cultural Dimensions

Chun Tao[1,2] · Alexander Glosenberg[3] · Terence J. G. Tracey[1] · David L. Blustein[4] · Lori L. Foster[5,6]

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Abstract

Are men and women more similar or different in their interests in careers? This question has propelled decades of research into the association between gender and vocational interests. However, our understanding of this question in an international context remains limited. In this study, we examined gender differences in vocational interests across national and cultural contexts by exploring whether national cultural dimensions would be associated with gender differences in the structure and mean levels of vocational interests in people/things, ideas/data, and prestige. Our findings support similarity in the structure of vocational interests for men and women across 42 countries based on two major models on interests. General trends of gender differences in interests emerge such that in comparison to men, women tend to report a large preference for working with people (versus things; $d = 1.04$), and smaller preferences for working with ideas (versus data; $d = 0.29$) and with prestige ($d = 0.18$). National cultural dimensions appear to moderate gender differences in interests beyond the influences of national gender inequality. Specifically, gender differences in interests in people (versus things) tend to be larger in countries of higher uncertainty avoidance and higher indulgence whereas gender differences in ideas (versus data) tend to be larger in countries of higher indulgence, uncertainty avoidance, and lower power distance. This study highlights how a better conceptualization of the influences of culture can inform vocational psychologists, gender studies researchers, and career counselors' work with men and women in understanding their vocational interests.

Keywords Vocational interests · Sex and gender measures · National cultural dimensions · Cross cultural test adaptation · RIASEC model · Octant model

Are men and women more similar or different in their interests in careers? This question has propelled decades of research into the association between gender and vocational interests (Lippa, 1998; Su & Rounds, 2015; Su et al., 2009; Tracey, 2002; Tracey & Caulum, 2015; Tracey & Robbins, 2005). To date, research has highlighted that women prefer to work with people rather than things (e.g., machines, tools); in contrast, men demonstrate a preference to work with things rather than people. Such differences in interests are large in magnitude and have been considered one of the most prominent gender differences observed in psychological research (Lippa, 1998; Lubinski, 2000; Su et al., 2009). Notably, vocational interests are associated with educational attainment, occupational choice, work performance, and

✉ Chun Tao
Tao.Chun@mayo.edu

Alexander Glosenberg
alexander.glosenberg@lmu.edu

Terence J. G. Tracey
Terence.Tracey@asu.edu

David L. Blustein
david.blustein@bc.edu

Lori L. Foster
lorifoster@ncsu.edu

1 Counseling and Counseling Psychology, Arizona State University, Tempe, AZ 85281, USA

2 Department of Psychiatry and Psychology, Mayo Clinic Arizona, Scottsdale, AZ 85259, USA

3 Department of Management, Loyola Marymount University, Los Angeles, CA 90045, USA

4 Department of Counseling, Developmental & Educational Psychology, Boston College, Chestnut Hill, MA 02467, USA

5 Department of Psychology, North Carolina State University, Raleigh, NC 27695, USA

6 School of Management Studies, University of Cape Town, Cape Town, South Africa

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work satisfaction (see, e.g., Morgan et al., 2001; Nye et al., 2012; Rounds & Su, 2014; Tracey & Robbins, 2005; Tracey et al., 2005). Therefore, understanding gender differences in interests may explain gender differences in occupational entry, especially with respect to science, technology, engineering, and mathematics (STEM) fields (see e.g., Ceci et al., 2014; Halpern, 2014; Lubinski et al., 2014; Schmidt, 2014; Tellhed et al., 2017).

However, one limitation to extant literature on vocational interests is a predominant focus on certain intra-national contexts, specifically on economically developed and individualistic countries like the United States (Morris, 2016). When cross-national gender differences in career-related aspirations, interests, or attainment have been explored, investigations often center on STEM-related interests (e.g., affinity for math) and on questions of economic development and gender equality/egalitarianism (Charles & Bradley, 2009; Connolly et al., 2020; Falk & Hermle, 2018; Ott-Holland et al., 2013; Stoet & Geary, 2018). Evidence has begun to accumulate showing a gender-equality "paradox" wherein gendered differences appear to be greater in societies of greater equality and/or where men and women tend to have greater economic resources (see e.g., Charles & Bradley, 2009; Stoet & Geary, 2018). In contrast to insights on career attainment in professions tied to such resource-based factors, relatively little is known regarding the extent to which differences in vocational interests might vary according to other cross-national factors, namely cultural dimensions (Hofstede et al., 2010). Despite a few recent studies that have begun to shed light on how cultural factors are related to vocational interests (see e.g., Falk & Hermle, 2018; Ott-Holland et al., 2013; Stoet & Geary, 2018), the interplay among culture, gender, and interests remains underexamined.

Cultural dimensions at the national level can affect individual-level experiences in unique ways. Yalcinkaya and Adams (2020) propose that cultural values might play a role in shaping career attitudes, goals, and choices. In fact, national-level power distance, individualism, masculinity, and uncertainty avoidance have been found to correlate with residents' personality traits (Hofstede & McCrae, 2004) and work-related behaviors (Taras et al., 2010). Therefore, we explored if gender differences in vocational interests would be associated with national cultural dimensions beyond the effects of national gender inequality in the present study. To this aim, we analyzed survey responses from a geographically diverse dataset of English-speaking adults collected on the website of an international English-language magazine, *Time*. To explore differences in vocational interests between men and women, we first tested the generalizability of two major models of vocational interests–the RIASEC (Holland, 1997) and the octant model (Tracey, 2002)–by gender across a broad range of countries. Following this, we compared the mean levels of interests between men and women across countries and investigated whether national cultural dimensions are related to these mean-level differences.

Brief Review of Vocational Interest Models

The development of major vocational interest theories has largely been grounded in the United States (see e.g., Holland, 1997; Prediger, 1982; Tracey & Rounds, 1995). One of the most influential and well-examined theories to date is Holland's (1997) theory of career choice. Holland proposes that individuals' vocational interests and work environments exist in an equilateral hexagonal plane. The six points of the hexagon represent six interest categories–realistic (R), investigative (I), artistic (A), social (S), enterprising (E), and conventional (C) interests. Interests that are more proximal to each other in the plane (e.g., R and I, I and A) more strongly correlate with each other than interests that are less so (e.g., R and S, I and E). Prediger (1982; see also Prediger & Vansickle, 1992) has posited that instead of a hexagonal structure, the RIASEC plane can be better characterized by two bipolar dimensions of interests – interests in people/things and ideas/data. Building upon these theories, Tracey and Rounds (1995) have found that interests are best portrayed as a circular structure. Further, they suggest that the number of vocational interest types (e.g., 6 or 8 or 4) is arbitrary. As such, Tracey (2002) has proposed an eight-type octant model to assess interests in social facilitating, managing, business detail, data processing, mechanical, nature/outdoors, artistic, and helping. The octant model has shown a better model fit than the RIASEC model (Tracey, 2002). Together, the six RIASEC types, the two Prediger's bipolar dimensions, and the eight octant types represent interests in the same two-dimensional circumplex plane that can be calculated interchangeably. Beyond a two-dimensional structure of interests, Tracey (2002; Tracey & Rounds, 1996) has identified interest in prestige as a third dimension orthogonal to the aforementioned plane, defining a spherical model. For this study, we focus on assessing interests in people/things, ideas/data, and prestige.

Gender Differences in Vocational Interests

There has been a longstanding history of investigations into factors related to gender differences and similarities (Eagly & Wood, 2013). Differences between men and women can be triggered or modified by socio-environmental conditions such as culture (Eagly & Wood, 2013; Schmidt, 2014). For example, gender role expectations may affect the ways boys and girls are treated (e.g., boys being provided trucks to play with and girls being provided with dolls to play with, boys

being encouraged while girls being discouraged to engage in certain activities). These gender-unique treatments may further influence boys' and girls' interests, personality, preferences, and self-identity in relation to gender norms (e.g., what boys and girls are expected to like and how they are expected to behave as they grow; see e.g., Blakemore, 1998), which may subsequently guide their occupational choices in adulthood.

To date, strong evidence has emerged in the literature on gendered vocational interests in people/things: women prefer to work with people whereas men favor working with things (see e.g., Su & Rounds, 2015; Su et al., 2009). Similarly, in the context of the RIASEC model, women and girls report greater artistic, social, and conventional interests whereas men and boys show stronger realistic and investigative interests (Morris, 2016; Su et al., 2009). Meta-analyses have shown that the magnitude of these gender differences is large, with effect sizes ranging from $d = .93$ (Su et al., 2009) to $d = 1.20$ (Lippa, 1998). Gender differences in interest in ideas compared to interest in data were mixed as reported by Morris (2016) who found a small preference for women to work with ideas using a U.S. sample ($d = .18$) whereas in an earlier meta-analysis of 81 samples, Su et al. (2015) found a small preference for women to work with data instead of ideas ($d = -.10$). Further, much less research has explored if gender is associated with interests in prestige (Tracey, 2002; Tracey & Rounds, 1996). Limited findings imply that the relation between gender and interests in prestige may be contingent upon career fields. For example, women working in stereotypical careers for their gender (i.e., people-oriented career fields) hold higher prestige status than men in similar fields (Goyder et al., 2003) whereas men in non-stereotypical careers (e.g., people-oriented careers) endorse stronger interests in prestige than men in traditional stereotypical careers for men (Dodson & Borders, 2006).

In summary, substantial evidence has shown large gender differences in interests in people/things; however, findings on interests in ideas/data and prestige are inconclusive. Thus, greater insight into gendered differences beyond the dimension of interests in people/things are needed. Furthermore, research to date has predominantly focused on intranational comparisons of gender differences in interests, such as the United States and Canada (see e.g., Goyder et al., 2003; Morris, 2016). Yet, gender gaps in educational and career choices, performance, and self-efficacy can vary considerably across nations (Stoet & Geary, 2018). Such cross-national variance indicates that vocational interests might vary in ways that are under-examined. Therefore, in this study, we aimed to conduct an international investigation to test whether vocational interests would differ by gender on three dimensions–people/things, ideas/data, and prestige.

The Interplay among Cultural Dimensions, Gender, and Vocational Interests

Research has paid growing attention to social and cultural influences on gender differences and similarities across nations (see e.g., Eagly & Wood, 2013; Lippa, 2010; Schmidt et al., 2007; Stoet & Geary, 2018). Notable theories regarding factors that affect such differences across nations often pertain to the interrelated effects of economic modernization and gender equality (Inglehart & Baker, 2000). Most research has adopted one of two frameworks. The first framework, known as the social role hypothesis (see e.g., Falk & Hermle, 2018), derives from social role theory (Eagly, 1987; Eagly & Wood, 1999) and its extension–role congruity theory (Eagly & Karau, 2002). These theories propose that people tend to hold gender-specific preferences instilled in them by society that allow them to fit into the gendered structure of society. Thus, individuals tend not to pursue careers incongruent with their gender roles and they tend to be more likely to be attracted to careers that fulfill their gender roles (e.g., women wanting to help people). The social role hypothesis integrates these observations with patterns of cultural change and economic modernization and predicts that stereotypical gender roles would become less prominent in societies with diminished gender inequality (Falk & Hermle, 2018). In countries with greater economic development and gender equality, progressive values (e.g., women can earn more money than their husband) are more encouraged and individuals may have more freedom to express their interests freely. This way, gender roles are less likely to be enforced or adhered to (Inglehart & Baker, 2000), and gender differences are expected to be smaller.

The second framework, known as the resource hypothesis (Falk & Hermle, 2018) suggests that greater resources related to greater economic development and greater gender equality enable individuals to make choices in line with gendered preferences. Thus, in countries with lower economic development and lower gender equity, fewer resources are available, which constrains individuals to adopt what work roles are accessible to them regardless of their preferences, resulting in smaller gender differences. In nations with greater economic development and gender equity, however, having more resources allows individuals to develop and determine interests more freely and diversely, resulting in greater gender differences. This seemingly surprising pattern–a tendency of greater gender differences in preferences among higher economic modernization and gender equality–has given rise to recognition of the gender-equality "paradox" (see e.g., Charles & Bradley, 2009; Stoet & Geary, 2018).

Recent cross-cultural studies have shed light regarding these two different perspectives (see e.g., Evans et al., 2002; Falk & Hermle, 2018; Lippa, 2010; Ott-Holland et al., 2013; Stoet & Geary, 2018). Some research has indicated greater gender differences in societies of greater economic modernization–in support of the resource hypothesis and the gender equality paradox (Falk & Hermle, 2018; Ott-Holland et al., 2013; Stoet & Geary, 2018). For example, Stoet and Geary (2018) examined gender differences in occupational preferences across 67 countries and found that countries with greater gender equality demonstrated greater gender differences. Falk and Hermle (2018) investigated preferences across 76 countries with a focus on those countries' Gross Domestic Product (GDP), and countries with greater GDP had greater gender differences. Another archival study across 20 countries (Ott-Holland et al., 2013) found that levels of societal gender egalitarianism were positively associated with gender differences in vocational interests. However, other research has indicated equivocal or more complex results (Charles & Bradley, 2009; Connolly et al., 2020; Lippa, 2010; Richardson et al., 2020). For example, one study across 53 countries (Lippa, 2010) found gender differences in vocational interests in people/things regardless of societal gender-egalitarian characteristics. Richardson et al. (2020) found that when different measures were used to assess gender equality, the significant positive association between national gender equality and gender differences observed by Stoet and Geary (2018) was found to be non-significant. This in part supports the possibility that gender equality is a multidimensional construct that may be associated with some cultural values and not others (Stefani & Prati, 2021). Charles and Bradley (2009) further argue that the relation between national economic development and gender differences is not linear and varies depending on a country's materialist and post-materialist contexts. Additionally, Connolly et al. (2020) conducted a study on the association between gender equality and values and found that this relation differs in direction when examined cross-sectionally versus longitudinally. These mixed results are important to recognize as they suggest that gender differences in interests may not be universal in their magnitudes and can be affected by the design and measures used in research.

Drawing upon equivocal findings on the role of economic modernization and gender equality, it is possible that additional insight into gender differences in interests may be gained through the consideration of cultural values. Yalcinkaya and Adams (2020) have proposed a cultural psychological model that posits cultural influences, such as the cultural values of individualism/collectivism, affect the goals and priorities related to individuals' attitudes and choices of career. Specifically, in post-materialist/more individualistic countries, men and women are more inclined to engage in self-expression, which may facilitate adherence to stereotypical gender identities and broaden gender differences, whereas in materialist/more collectivistic countries, men and women may focus more on financial security, so more women may pursue STEM-related fields that provide greater economic security, leading to smaller gender differences. The cultural psychological model (Yalcinkaya & Adams, 2020) adds to the resource hypothesis by highlighting that in self-expressive societies, academic and career goals/choices tend to be made by individuals toward the perceived 'authentic' expression of their masculine and/or feminine identities. This tendency then leads to more pronounced discrepancies in academic and career attainment between men and women. Despite the greater explanatory potential of the cultural psychological model, its discussion of culture is limited predominately to the influences of individualism and collectivism. As discussed below, a broader appreciation of the relations of additional cultural dimensions to gender differences in interests is needed.

In the cultural value framework, Hofstede (1984) has proposed a set of six dimensions to represent how national cultures vary. Specifically, these six dimensions include individualism (versus collectivism), power distance, uncertainty avoidance, masculinity (versus femininity), long-term (versus short-term) orientation, and indulgence (versus restraint). Individualism reflects societal values favoring individual actions rather than actions as a member of groups. Power distance represents societal tolerance and acceptance toward unequal distributions of power. Uncertainty avoidance indicates societal avoidance of uncertain situations by establishing formal regulations and enhancing occupational stability. Masculinity symbolizes societal preferences of stereotypical values for men like assertiveness over stereotypical values for women like cooperation. Long-term orientation denotes societal preferences of future-oriented values like persistence and thrift in comparison to present- and past-oriented values like honoring the traditions. Additionally, indulgence characterizes collective preferences of following human drives toward joy rather than regulation and suppression of such drives. Individualism, power distance, uncertainty avoidance, and masculinity at the national level have been widely researched and have demonstrated moderate correlations with individuals' personality traits (Hofstede & McCrae, 2004). These four national cultural dimensions have also shown small to moderate associations with work-related behaviors such as communication or conflict resolution (for a review, see Taras et al., 2010).

Based upon the potential for cultural values to shape individuals' vocational interests, goals, and priorities, we reasoned that national cultural dimensions would be associated with gender differences in interests over and above the frequently examined influences of national gender equality. Furthermore, in accordance with Yalcinkaya and Adams' (2020) conceptualization, we hypothesized that cultural values tied to a greater tendency toward individual

self-expression would be associated with greater gender differences, and cultural values consistent with a greater emphasis on financial security would be associated with smaller gender differences. Following this logic, not only might gender differences be affected by national individualism, but they might vary according to Hofstede's (1984) other five cultural dimensions. Therefore, we formed four hypotheses and considered whether and how national long-term orientation and indulgence might be related to gender differences in interests as exploratory questions given limited evidence of their influences in prior research.

First, we hypothesized that countries that favor values of individualism would be expected to show greater gender differences in interests. We based this hypothesis on the cultural psychological model (Yalcinkaya & Adams, 2020) which predicts that countries with a greater focus on individualism (i.e., greater national individualism) would demonstrate an enhanced priority for self-expression in accordance with individuals' own goals and preferences and less emphasis on financial security. Together, these tendencies could encourage men and women to express their genuine gendered identities, which would be linked to broader gender differences in interests. Second, we hypothesized that nations with lower national power distance would demonstrate greater gender differences in interests. That is, in countries endorsing lower power distance (i.e., people are more equal), individuals would be less constrained, and they could express their personal interests in a way that is more congruent with their authentic selves (Yalcinkaya & Adams, 2020), which would be associated with greater gender differences in interests. The relatively lower constraints on interests in countries with lower power distance would be similar to previous findings on the gender-equality "paradox" in countries with greater economic development and gender equality (see e.g., Charles & Bradley, 2009; Stoet & Geary, 2018). Third, we hypothesized that greater national masculinity would be associated with greater gender differences in interests. Countries endorsing greater national masculine values (e.g., assertiveness & valuing money) may endorse lower feminine values (e.g., agreeableness, cooperation); these national values likely encourage a more self-assertive individual expression that is associated with greater gender differences in vocational interests in countries higher in masculine values. Fourth, because countries higher in uncertainty avoidance may be more rule-governed and stability-oriented, individuals may be constrained to pursue their own self-expression which might run counter to societal rules and group stability. Instead individuals would be expected to place a greater focus on financial security to maintain stability in society, which would likely be associated with more similar vocational interests between men and women; therefore, we expected smaller gender differences in interests being associated with higher national uncertainty avoidance.

The Importance of the Structure of Interests

Cross-cultural investigations of gender differences rely heavily on the equivalence of measures across both different countries and gender. To facilitate meaningful comparisons of mean scores across countries, measures ought to function similarly in each country and between men and women. In the context of the RIASEC and the octant models of vocational interests, if scores do not adhere to the theoretical circumplex structure for both men and women across countries, they may be measuring different constructs. That is, the prerequisite for a mean comparison of interest scores by gender and countries is a common structure of interests. If responses from one country do not adhere to the common theoretical circumplex model, the circumplex model may not be a valid representation of interests for individuals in that country. In other words, people in that country may think about interests differently than individuals in other countries whose responses conform to the common model. This examination, depending on adherence to the common interest structure, provides critical implications for how interests are constructed across countries. Rounds and Tracey (1996) adopted such an approach in their meta-analysis of the validity of the RIASEC circumplex structure across 19 countries. They found relatively little evidence for the circumplex structure of interests in countries besides the United States. Furthermore, neither GDP nor individualism at the national level was related to model fit. In light of these findings, an important research priority is to examine how national cultural dimensions may be related to the structure of interests by gender across countries.

The Present Study

This study aimed to examine gender differences in the structure and mean scores of vocational interests in an international context and expand our understanding of the effects of national cultural dimensions. There were three overarching research questions. First, we examined the generalizability of the structures of the RIASEC (Holland, 1997) and the octant (Tracey, 2002) models by gender across nations of diverse cultural backgrounds to determine the appropriateness of mean score comparisons. We used publicly available data from a mini version of the Personal Globe Inventory (Tracey, 2002) administered by *Time* magazine in English to readers of its website worldwide. The PGI-Mini has demonstrated superior fit to both interest models compared to existing measures developed from the RIASEC model in the United States and international contexts (Glosenberg et al., 2019). The utilization of data from an English survey was justified given the global influence and central linguistic connections of English to other languages (Ronena et al.,

2014). We also explored cultural effects on the structure of interests by probing the interactions between national cultural dimensions and gender on model fit indices.

Second, among nations in which the three-dimensional vocational interest structure was functionally invariant across gender, we examined whether men and women would differ in their interests in people/things, ideas/data, and prestige. In line with past research, we expected that women would be more likely to prefer to work with people (versus things) than men and the effect size of this difference would be large (Su & Rounds, 2015). Potential gender differences in ideas/data and prestige would be explored.

Finally, we investigated the interplay between gender and national cultural dimensions on vocational interests. We hypothesized that power distance, individualism, masculinity, and uncertainty avoidance would moderate dimensions of vocational interests beyond the effects of national gender inequality. Specifically, we expected lower power distance, greater individualism, greater masculinity, and lower uncertainty avoidance at the national level would broaden gender differences in interests. With limited prior evidence, we explored potential moderating effects of national long-term orientation and indulgence.

Method

Participants and Procedure

Data used in this study were drawn from a large publicly accessible dataset – specifically, data on vocational interests were drawn from an online survey conducted by *Time*, an internationally popular English-language news and lifestyle magazine. The initial dataset included 84,394 responses (18 years or older) from 193 nations/territories/regions. For consistency, these regions are referred to as countries or nations hereafter. Responses that did not indicate the respondents' gender and those who reported being older than 99 years old were first excluded. Given that consecutive items on the PGI-Mini (Tracey, 2021) always assessed different dimensions of interests, responses with multiple consecutive identical responses were likely to have been generated carelessly and they were removed. We retained data from countries with at least 30 responses tied to men and 30 responses tied to women to allow adequate examination of gender differences in interest models and generalizability across nations. This resulted in a final dataset of 78,160 responses from 59 countries. The average age reported was 37.92 years old (*SD* = 13.51), ranging from 18 to 99 years old. Most responses were tied to individuals having at least a bachelor's degree (n = 31,580, 40.40%) and 37.77% reported having a postgraduate degree (n = 29,521), whereas 10.63% reported a high school diploma or GED as their highest degree (n = 8,307), and 7.96% reported their highest education as less than high school or still in high school (n = 622), with 3.24% missing education information. Approximately three quarters of responses were tied to individuals who were employed (n = 58,768, 75.19%). Although income levels were reported, these values were in dollars – making interpretation of them challenging internationally. Still, about half of responses were tied to individuals with an annual income below $99,999, including 28.83% reporting an annual income below $49,999 ($n$ = 22,534) and 28.86% between $50,000 and $99,999 ($n$ = 22,557), whereas 11.71% (n = 9,154) and 7.30% (n = 5,704) reported an annual income in the $100,000-$149,999 range and higher than $150,000, respectively, and the rest (23.30%) had missing information on income. The distribution of responses across gender and nations can be found in Table 1.

The original online survey conducted by *Time* was advertised as a way for respondents to explore their "work personalities" and how those personalities matched with their jobs. Respondents reported demographic information including age, gender, education, employment status, and job title if employed. Following this, respondents completed the 20-item Personal Globe Inventory-Mini (PGI-Mini; Tracey, 2021). The second and fifth author's university Institutional Review Board approved the use of the publicly available archival data in the present study (https://doi.org/10.17605/OSF.IO/BE5JA).

Measures

Vocational Interests

Respondents reported their vocational interests on the Personal Globe Inventory-Mini (PGI-Mini; Tracey, 2021), a shortened version of the Personal Globe Inventory-Short (PGI-Short; Tracey, 2010) and the original Personal Globe Inventory (PGI; Tracey, 2002). The PGI-Mini administers 20 items selected from the PGI based upon an item response theory analysis of the discrimination each item exhibited in relation to the respective PGI construct (Tracey, 2010). The PGI-Mini assesses individuals' interests in various activities, such as "overseeing a hotel", "preparing financial reports", and "helping children with learning problems" on a 7-point Likert scale ranging from 1 (*strongly dislike*) to 7 (*strongly like*). The PGI-Mini, similar to the PGI-Short and the PGI, assesses vocational interests on three dimensions: interests in people/things, ideas/data, and prestige (Tracey & Rounds, 1996). The PGI-Mini yields scores on both the RIASEC and the octant scales. Aggregate scores were calculated for the people/things and ideas/data bipolar dimensions and prestige, with higher scores reflecting greater interests in people (versus things), ideas (versus data), and prestige (Tracey,

Table 1 Sample Sizes, Correspondence Indices (CIs), Fit Probabilities, and Mean Scores on People/Things, Ideas/Data, and Prestige Dimensions by Gender Along with the Effect Size (*ES*, Cohen's *d*)

Country	Responses		Age (in Years)		*CI*: Octant				*CI*: RIASEC				People/Things			Ideas/Data			Prestige		
	Men	Women	Men	Women	Men	*p*	Women	*p*	Men	*p*	Women	*p*	Men	Women	*ES*	Men	Women	*ES*	Men	Women	*ES*
United States	15021	27452	41.47	39.19	.87	.000	.82	.000	.86	.017	.89	.017	-0.94	3.59	0.90	-0.08	1.24	0.23	1.78	1.96	0.14
Netherlands	2127	1687	40.88	36.92	.78	.000	.80	.000	.75	.017	.89	.017	-0.75	4.23	1.04	0.09	2.11	0.38	1.02	1.59	0.41
Canada	1326	3403	39.03	36.79	.79	.000	.77	.000	.83	.017	.78	.033	-1.15	3.42	0.94	-0.68	0.41	0.20	1.61	1.59	0.02
United Kingdom	1292	1811	38.34	33.92	.78	.000	.77	.000	.83	.017	.89	.017	-0.78	3.64	0.93	0.39	1.68	0.23	1.84	2.00	0.12
Australia	863	1704	34.97	33.22	.81	.000	.78	.000	.81	.017	.86	.017	-0.39	3.73	0.81	-0.60	1.19	0.32	1.63	1.74	0.08
India	765	501	30.95	28.95	.83	.000	.74	.000	.81	.017	.89	.017	0.85	4.44	0.75	-0.50	0.80	0.24	2.04	2.25	0.16
South Africa	560	607	42.37	39.06	.84	.000	.76	.000	.75	.017	.78	.017	-1.26	3.36	0.92	0.16	1.47	0.24	2.00	2.15	0.13
Philippines	488	1158	33.15	29.91	.87	.000	.85	.000	.92	.017	.86	.017	0.59	3.66	0.61	-0.93	-1.15	0.04	2.00	2.11	0.08
Germany	461	620	37.83	35.09	.73	.000	.87	.000	.64	.017	.83	.017	-0.36	3.20	0.76	0.14	1.88	0.32	2.14	2.35	0.15
Malaysia	398	643	32.48	29.41	.70	.000	.81	.000	.72	.017	.86	.017	0.37	3.56	0.68	-0.62	-0.25	0.07	1.79	1.50	0.23
Indonesia	341	665	29.89	28.09	.79	.000	.76	.000	.72	.017	.83	.017	0.75	3.46	0.58	-0.99	-0.86	0.03	1.78	1.89	0.09
Italy	317	406	39.77	36.90	.71	.000	.81	.000	.64	.017	.86	.017	-0.88	3.85	0.90	0.20	1.49	0.23	1.96	2.35	0.30
Singapore	317	704	34.09	32.32	.75	.000	.82	.000	.78	.033	.86	.017	0.86	3.94	0.65	-1.12	0.53	0.33	1.93	1.57	0.29
France	303	480	39.46	36.30	.77	.000	.72	.000	.81	.017	.81	.017	0.54	4.04	0.75	0.17	2.15	0.36	2.28	2.25	0.02
Mexico	302	360	38.48	33.96	.82	.000	.75	.000	.81	.017	.81	.017	-0.18	4.32	0.90	0.45	1.82	0.23	2.20	2.46	0.21
Spain	254	349	39.53	35.34	.67	.000	.77	.000	.72	.017	.67	.017	0.44	5.14	0.99	0.32	1.52	0.21	2.25	2.44	0.15
Ireland	253	346	38.13	35.42	.85	.000	.82	.000	.89	.017	.92	.017	-0.49	3.56	0.79	0.02	1.36	0.22	1.61	2.04	0.32
Hong Kong	200	461	35.14	32.35	.68	.000	.69	.000	.89	.017	.78	.017	0.55	4.07	0.79	-0.48	0.27	0.15	1.59	1.76	0.13
Brazil	197	185	35.40	33.18	.73	.000	.72	.000	.76	.017	.75	.017	0.33	4.38	0.78	-0.77	1.25	0.37	2.14	2.22	0.05
New Zealand	180	305	35.97	34.00	.90	.000	.77	.000	.83	.017	.75	.017	0.38	3.23	0.62	0.01	1.50	0.28	1.67	1.85	0.14
United Arab Emirates	175	219	35.82	32.34	.74	.000	.80	.000	.89	.017	.97	.017	1.05	4.74	0.74	-1.15	0.16	0.23	1.78	1.95	0.12
Sweden	172	203	41.07	37.18	.75	.000	.76	.000	.60	.017	.86	.017	-0.91	3.26	0.85	0.88	1.33	0.09	2.02	2.63	0.47
Belgium	171	241	38.84	32.64	.65	.000	.62	.000	.67	.017	.81	.017	-0.76	4.18	1.11	-0.09	1.41	0.27	1.84	2.40	0.41
Turkey	169	166	33.43	32.93	.64	.000	.76	.000	.60	.033	.75	.017	-0.15	4.84	1.02	-0.37	2.57	0.52	1.94	2.33	0.28
Japan	165	220	38.47	36.38	.78	.000	.67	.000	.81	.017	.58	.017	-0.13	3.27	0.72	-0.11	1.49	0.31	1.69	1.99	0.24
Portugal	164	252	35.49	33.29	.76	.000	.67	.000	.75	.017	.67	.017	-0.50	3.77	0.89	-0.16	1.74	0.33	1.97	2.29	0.25
Israel	161	186	46.02	43.66	.72	.000	.65	.000	.72	.017	.69	.017	-0.43	3.54	0.86	0.86	2.72	0.32	2.36	2.47	0.08
Greece	153	190	38.75	34.66	.66	.000	.70	.000	.67	.017	.61	.017	-0.59	4.36	1.02	-0.56	2.69	0.61	2.25	2.14	0.08
Colombia	148	147	35.38	31.16	.70	.001	.67	.000	.67	.033	.69	.017	0.55	4.94	0.95	0.46	1.00	0.09	2.23	2.65	0.33
Pakistan	144	95	29.76	27.49	.59	.000	.52	.001	.72	.017	.67	.017	1.03	4.40	0.68	-2.77	0.56	0.69	2.02	1.88	0.10
Switzerland	144	239	39.69	38.17	.69	.000	.72	.000	.68	.017	.67	.017	-0.46	4.09	0.95	-0.17	1.40	0.30	2.18	2.36	0.13
Norway	124	187	41.73	36.34	.63	.001	.78	.000	.53	.017	.72	.017	-1.13	3.53	1.09	0.01	2.80	0.52	1.92	2.63	0.57
Thailand	123	143	37.36	33.27	.54	.000	.79	.001	.61	.033	.89	.017	-0.08	3.51	0.72	-0.96	-0.60	0.07	1.46	1.88	0.31
Saudi Arabia [ex]	115	70	36.69	32.94	.59	.000	.54	.000	.50	.017	.67	.017	-1.12	3.20	1.00	-1.90	0.37	0.44	1.83	1.91	0.06
Nigeria [ex]	106	53	34.86	33.87	.54	.000	.43	.001	.50	.067	.58	.033	-0.46	3.19	0.70	-2.80	-2.50	0.06	2.67	2.39	0.23
Republic of Korea	99	150	37.60	31.82	.73	.000	.77	.000	.69	.017	.75	.017	0.39	4.25	0.87	-1.43	0.80	0.45	2.01	2.30	0.22

Table 1 (continued)

Country	Responses		Age (in Years)		*CI*: Octant				*CI*: RIASEC				People/Things			Ideas/Data			Prestige		
	Men	Women	Men	Women	Men	*p*	Women	*p*	Men	*p*	Women	*p*	Men	Women	*ES*	Men	Women	*ES*	Men	Women	*ES*
Finland	96	132	41.30	37.36	.69	.000	.54	.001	.67	.017	.64	.033	-0.73	3.26	0.81	0.15	2.66	0.46	1.93	2.32	0.30
Denmark [ex]	95	173	37.37	35.28	.67	.000	.60	.002	.58	.033	.47	.083	0.19	4.05	0.85	-0.86	2.57	0.64	2.26	2.66	0.36
Taiwan [ex]	84	192	33.11	31.21	.56	.001	.66	.000	.25	.183	.75	.017	0.30	4.02	0.91	-0.57	0.36	0.21	2.21	1.90	0.21
Romania	79	149	28.03	30.24	.51	.000	.73	.000	.57	.017	.75	.017	0.12	3.90	0.74	0.06	2.46	0.45	2.08	2.10	0.02
Poland	73	154	32.74	30.21	.72	.000	.65	.001	.72	.017	.72	.017	-0.24	3.62	0.76	1.20	0.95	0.05	1.73	1.99	0.17
Hungary	72	130	36.88	34.45	.64	.000	.60	.001	.64	.033	.69	.017	-0.08	3.98	0.77	0.78	2.22	0.27	1.85	2.69	0.68
Vietnam [ex]	71	79	30.72	29.46	.43	.003	.58	.000	.33	.150	.63	.017	0.54	5.72	1.33	-0.25	0.88	0.22	2.34	2.41	0.06
Chile	69	55	36.43	33.38	.77	.000	.61	.001	.69	.017	.53	.033	1.64	4.33	0.58	-0.25	2.93	0.56	2.49	2.13	0.30
Austria [ex]	68	119	36.57	33.68	.47	.004	.61	.000	.53	.017	.69	.017	0.13	3.27	0.62	-1.02	1.54	0.46	2.66	2.34	0.25
Argentina [ex]	62	88	37.97	34.64	.53	.003	.66	.001	.44	.083	.50	.033	-1.14	4.87	1.22	-0.67	1.88	0.55	2.06	2.74	0.49
Georgia	61	123	30.72	31.23	.63	.001	.60	.002	.56	.033	.58	.033	0.20	2.78	0.51	-0.03	0.23	0.05	2.25	2.71	0.36
Costa Rica [ex]	51	60	34.59	35.12	.72	.000	.41	.003	.78	.017	.50	.067	0.93	4.22	0.66	-1.12	1.85	0.52	2.07	2.04	0.02
Egypt [ex]	49	47	30.82	29.83	.66	.000	.50	.000	.69	.033	.35	.033	-0.86	3.73	0.98	-0.79	2.18	0.51	1.43	1.93	0.33
Lebanon [ex]	45	45	36.82	31.33	.42	.002	.47	.000	.47	.033	.61	.017	-0.99	3.90	1.03	-0.17	1.70	0.34	2.37	2.36	0.01
Puerto Rico [ex]	45	66	44.33	39.18	.61	.001	.21	.066	.78	.017	.36	.033	0.21	4.12	0.78	-1.29	0.54	0.34	2.21	2.30	0.07
Kenya [ex]	42	35	34.05	31.80	.61	.000	.62	.002	.53	.067	.69	.033	-0.50	4.26	0.87	-2.08	0.31	0.43	1.90	2.17	0.16
Venezuela	41	34	42.66	36.76	.62	.000	.55	.000	.67	.017	.58	.033	-1.87	5.77	1.49	-0.56	0.65	0.24	2.02	2.75	0.47
Russia [ex]	40	57	35.78	31.33	.40	.012	.50	.004	.56	.050	.39	.117	0.10	2.93	0.60	-0.24	2.37	0.48	1.78	2.06	0.24
Peru [ex]	38	39	36.00	30.92	.48	.002	.49	.001	.64	.017	.44	.050	0.88	3.45	0.51	0.25	0.13	0.03	2.07	2.74	0.43
Bulgaria	36	44	28.39	29.95	.64	.000	.71	.000	.61	.017	.53	.033	-0.46	3.76	0.87	0.56	2.84	0.43	1.97	2.06	0.05
Ecuador [ex]	36	32	37.17	34.81	.48	.001	.42	.011	.61	.017	.78	.017	0.42	4.56	0.82	0.39	0.78	0.06	2.57	2.17	0.33
Czech Republic [ex]	33	45	36.76	32.69	.47	.008	.48	.007	.53	.033	.38	.100	0.15	3.65	0.73	-1.41	0.40	0.30	1.89	2.59	0.44
Qatar [ex]	32	38	40.09	33.47	.30	.028	.53	.003	.28	.183	.42	.067	-0.64	3.34	0.96	-1.81	0.82	0.56	1.59	2.20	0.42
Mean	-	-	39.41	37.01	0.67	-	0.66	-	0.67	-	0.70	-	-0.12	3.92	0.84	-0.42	1.21	0.31	1.99	2.20	0.23
SD	-	-	14.19	13.00	0.13	-	0.13	-	0.15	-	0.16	-	0.72	0.62	0.19	0.83	1.08	0.17	0.31	0.32	0.15
Mean non excluded	-	-	39.54	37.11	0.73	-	0.73	-	0.73	-	0.76	-	-0.04	3.96	0.84	-0.37	1.25	0.30	2.03	2.23	0.21
SD non excluded	-	-	14.24	13.04	0.09	-	0.09	-	0.10	-	0.11	-	0.72	0.67	0.20	0.80	1.13	0.17	0.29	0.28	0.15

Sorted by octant CI among men's responses

ES effect size as in *Cohen's d*

[ex] denotes countries that were excluded from random effects meta-analysis on the interactional effects of cultural dimensions and gender on interests

2010). The PGI-Short and the PGI have demonstrated satisfactory fit to the RIASEC and octant models in the United States (Tracey, 2002, 2010) and some international contexts (for a review, see Tao et al., 2019). In addition, the PGI-Short has shown evidence of satisfactory two-week test–retest reliability and little differential item functioning across gender and ethnicity with a U.S. sample (Tracey, 2010). In this study, the mean internal consistency estimates across all 59 countries were adequate but not high, given the brief nature of the instrument: people/things mean α = .70 (*SD* = .07, range = .35–.90), ideas/data mean α = .70 (*SD* = .09; range = .31–.89), and prestige mean α = .79 (*SD* = .06, range = .58–.95).

National Gender Inequality

National gender inequality was estimated using the United Nations Development Programme's measure of gender inequality (2020) as estimated by country-level factors (measures of education, health, labor force participation, and political participation regarding gendered shares of parliamentary seats) in 2016. The index ranges from 0 to 1.00 and higher values indicate greater gender inequality at the national level.

National Cultural Dimensions

National cultural dimensions were obtained from Hofstede and Minkov's (2013) updates to Hofstede et al.'s (2010) scores for the respective nations included in the final dataset. The Hofstede et al.' scores (2010) are widely used measures of workplace values at the national level (see Taras et al., 2010). The updated dimensions include power distance, individualism (versus collectivism), masculinity (versus femininity), uncertainty avoidance, long-term (versus short-term) orientation, and indulgence (versus restraint). Index scores range from 0 to 100. A higher cultural dimension index score indicates that a country as a whole endorses more strongly the respective cultural dimension in comparison to other countries.

Demographics

Respondents' self-identified gender was coded (men = 0.5, women = -0.5). We used Internet protocol (IP) addresses tied to responses to determine respondents' country of residence at the time of survey completion.

Data Analysis

To examine the validity of models of vocational interests between men and women across countries, we calculated correspondence index (CI) values according to two popular models – Holland's (1997) RIASEC model and Tracey's (2002) octant model. CI values are based on the assumption that estimates of different interests should exhibit higher correlations with more related interests, and lower correlations with less related interests (Tracey, 1997). CI values (similar to Somer's *D*) are calculated using randomization tests of hypothesized relations among interests using the RTHORR package in R (Tracey, 1997). Results from this analysis yield a CI value and a probability of the extent to which data deviate from a model – in comparison to what deviation might be expected by chance. A CI value of .50 indicates that 75% of the predictions are met and 25% violated. Larger samples tend to yield stable matrices whereas smaller samples yield less stable matrices. To provide an indication of the precision of each CI, we computed the 95% confidence band around each index, using 1,000 bootstrapped responses from the data from each country. Thus, for each country, we obtained the significance of the data fit to a circumplex, the CI of this fit, and its 95% confidence band.

Following randomization tests, we performed hierarchical regression analyses using SPSS 27.0 based on the national-level data to examine whether there were gender differences in the model fit for the octant or the RIASEC structure, and whether the potential gender differences in model fit were moderated by national cultural dimensions. For each test on model fit, one model was performed for the moderating effects of each of the six cultural dimensions.

To examine gendered differences along interests in people/things, ideas/data, and prestige dimensions, a sufficient level of model validity is required. Thus, following an analysis of country-level factors with CI values, we excluded nations without adequate fit to either the octant or the RIASEC model among men or women (CI < .50 or $p > .05$; Tracey, 1997). We then estimated differences in men and women along all three interest dimensions with *t*-tests – yielding Cohen's *d* effect size estimates. Given that responses ranged widely across countries (68 in Ecuador to 42,473 in the United States; $M = 1{,}325$, $SD = 5{,}476$), we corrected for the effect of sampling error in our analysis of gender differences in interests in people/things, ideas/data, and prestige; in addition, because the reliability of the PGI-Mini varied across countries, we corrected for measurement error in our effect size estimates. These corrections were made using procedures from Schmidt and Hunter (2015). We then estimated differences in men and women along all three interest dimensions with *t*-tests – yielding Cohen's *d* effect size estimates.

To explore the relationship between national cultural dimensions and gendered differences in interests beyond the effects of national gender inequality, we conducted meta-analytic regression utilizing mixed-effect models with national gender inequality entered in the first step and national gender inequality and all six national cultural

dimensions entered in the second step of the model. Specifically, we utilized procedures from Cheung (2015) with estimates of standardized and non-standardized coefficients from Mplus Version 8 and estimates of variance explained by models from metaSEM in R.

Results

Vocational Interest Structures by Gender and Nations

Randomization tests of hypothesized order relations for both the octant and RIASEC models by gender and nations based on 59 countries can be found in Table 1. Most countries demonstrated adequate fit with CI values of .50 or higher at a significance level of $p < .05$. Specifically, based on the octant model, 86% of the CI values across gender and nation were greater than .50, which reflected that 86% of the countries showed a model fit meeting 75% of hypothesized order predictions. Furthermore, Rounds and Tracey (1996) set the CI value benchmark in the United States at .70. In the present study, 48% of the CI values were greater than .70. Compared to responses from men and women in the United States, interests reported by men in the Philippines and New Zealand and women in the Philippines, Germany, and Singapore showed similarly good or better fit to the octant model. Based on the RIASEC model, 89% of the CI values for either men or women were higher than .50, and 48% surpassed the U.S. benchmark of .70. Additionally, CI values for men in the Philippines, Ireland, Hong Kong, and United Arab Emirates, and women in the Netherlands, United Kingdom, India, Ireland, and United Arab Emirates were equivalent to or higher than their counterparts in the United States. Illustrations of the CI values with 95% confidence intervals for the octant and RIASEC models can be found in Figs. 1 and 2, respectively.

A paired-samples *t* test demonstrated that the overall fit to the RIASEC model across gender ($M = .68$, $SD = .15$) was better than the fit to the octant model ($M = .67$, $SD = .13$), $t(117) = 1.98$, $p = .05$, although the effect size of this difference was small ($d = 0.07$). Hierarchical regression analyses did not reveal any gender differences in fit to either model (see Table 2), nor did we find any significant interaction between gender and cultural dimensions on model fit.

Gender Differences in Interests

Descriptive statistics of interests in people/things, ideas/data, and prestige between men and women as well as effect sizes of gender differences by gender are summarized in Table 1.

Based on a winnowed dataset of 75,908 responses from 42 countries with adequate model fit, for interests in people/things, women showed a general trend toward more interest in working with people rather than things in comparison to men. A distribution of effect sizes for each country, ranging from 0.63 to 1.74, is depicted in a forest plot (see Fig. 3). After correcting for sampling and measurement error, the overall effect size of gender differences in interests in people/things across countries was large (Cohen, 1988), $d = 1.04$, 95% CI[1.01, 1.07], but there was a meaningful amount of heterogeneity in effect sizes as measured by a significant Q-statistic (180.45, $df = 41$, $p < .001$) and large I^2 statistic (77.28), ranging from an effect size of 0.63 in Georgia to 1.74 in Venezuela for interests in people/things.

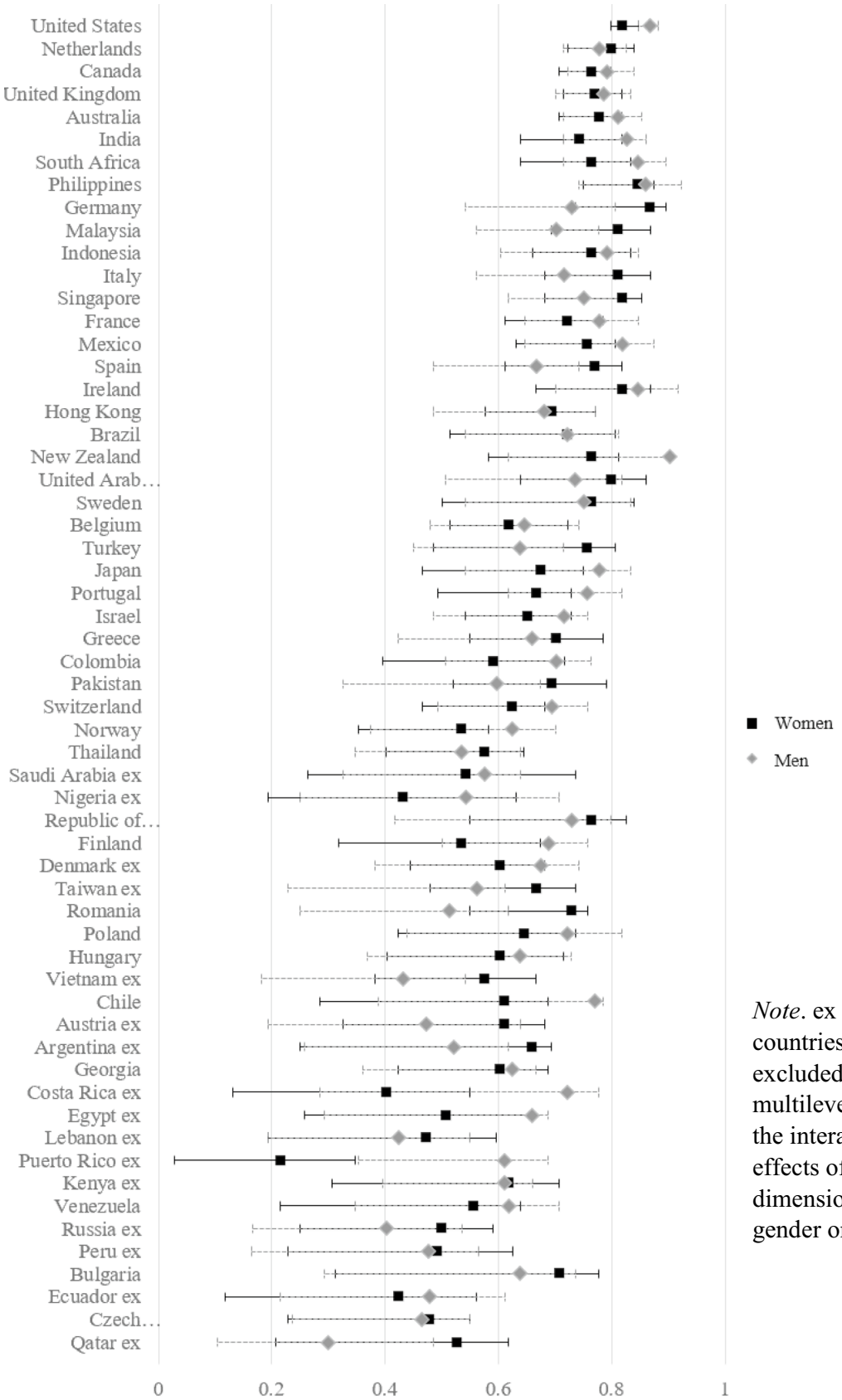
For interests in ideas/data, women tended to be more interested in working with ideas (versus data) than men in all but two countries (the Philippines and Poland). A forest plot of effect sizes for each country, ranging from 0.03 to 0.80, can be found in Fig. 4. After correcting for sampling and measurement error, the overall effect of interest differences between men and women across countries was small (Cohen, 1988), $d = 0.29$, 95% CI [0.26, 0.31], but there was a meaningful amount of heterogeneity in effect sizes as measured by a significant Q-statistic (137.01, $df = 41$, $p < .001$) and a somewhat large I^2 statistic (70.08), ranging from an effect size of 0.03 for Indonesia to 0.80 in Pakistan for interests in ideas/data.

For interests in prestige, women tended to be more interested in prestige than men in all but eight countries (Canada, Chile, France, Greece, Malaysia, Pakistan, Republic of Korea, and Singapore). A forest plot of effect sizes for each country, ranging from 0.02 to 0.78, can be found in Fig. 5. After correcting for sampling and measurement error, the overall effect between genders across countries was very small $d = 0.18$, 95% CI [0.15, 0.21], but a meaningful amount of heterogeneity in effect sizes was observed as evidenced by a significant Q-statistic (170.66, $df = 41$, $p < .001$) and large I^2 statistic (75.98), ranging from an effect size of 0.02 in France to 0.78 in Hungry for interests in prestige.

It may be helpful to note that the gender difference in interests in people/things ($d = 1.04$) appears to be over three times (3.58) the gender difference in interests in ideas/data ($d = 0.29$) and over five times (5.78) the gender difference in interests in prestige ($d = 0.18$).

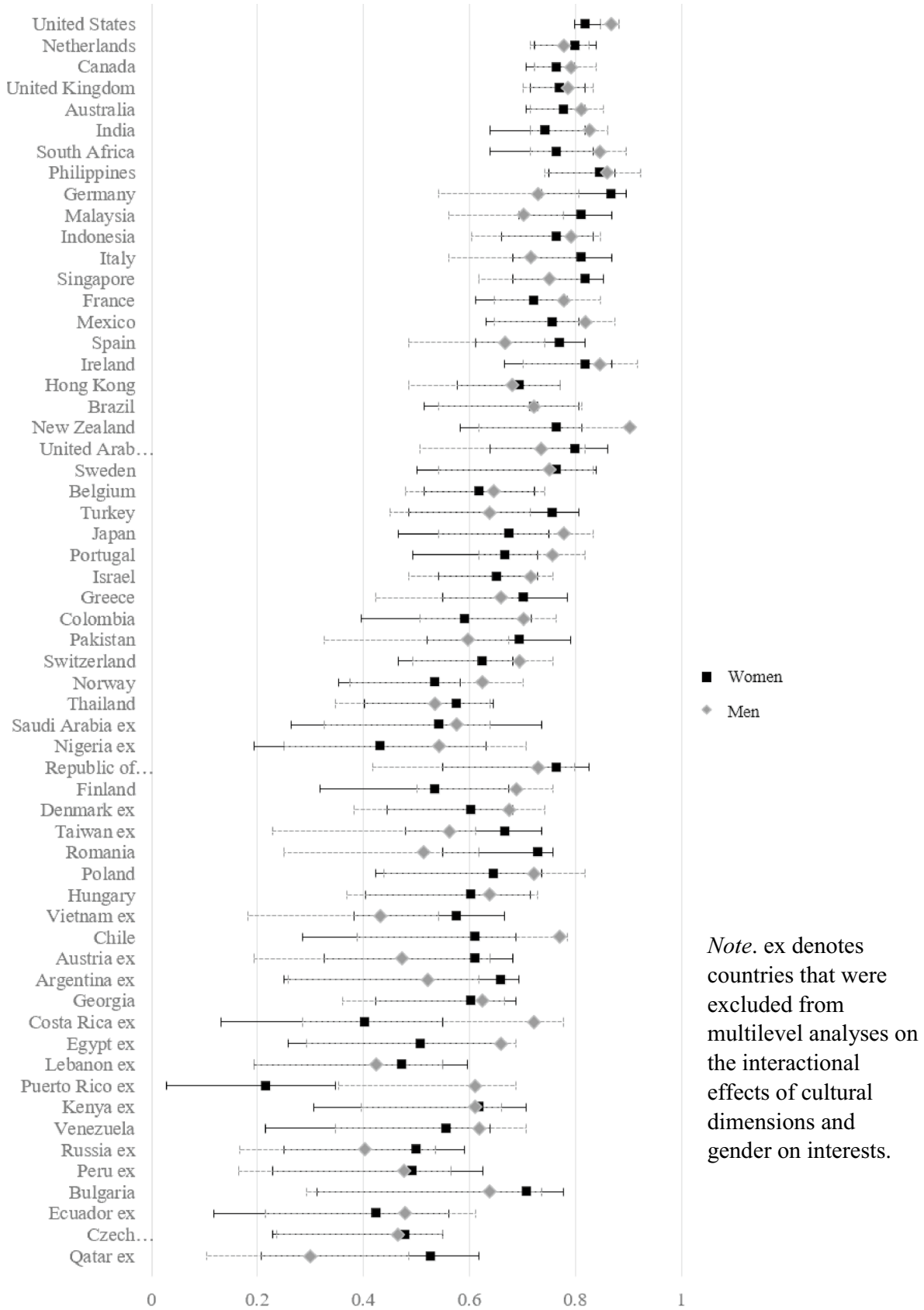
National Cultural Dimensions as Moderators

Results from the meta-analytic regression models for each of the three interest dimensions can be found in Table 3. For interests in people/things, national gender inequality explained 24% of heterogeneity across countries in effect



Note. ex denotes countries that were excluded from multilevel analyses on the interactional effects of cultural dimensions and gender on interests.

Fig. 1 Forest Plot of Correspondence Indices with 95% Confidence Intervals for Fit to the Octant Model for Women and Men by Country



Note. ex denotes countries that were excluded from multilevel analyses on the interactional effects of cultural dimensions and gender on interests.

Fig. 2 Forest Plot of Correspondence Indices with 95% Confidence Intervals for Fit to the RIASEC Model for Women and Men by Country

Table 2 Hierarchical Regression Models with Model Fit as Criteria with Country-Level Data

	Octant Model Fit											
	PDI		IDV		PDI		UAI		LTO		IND	
	b/β	*SE*	*b*/β	*SE*	*b*/β	*SE*	*b*/β	*SE*	*b*/β	*SE*	*b*/β	*SE*
Step 1												
Gender[a]	0.01/0.03	0.02	0.01/0.03	0.02	0.01/0.03	0.02	0.01/0.03	0.02	0.002/0.01	0.02	0.001/0.003	0.02
Culture[b]	-0.001/-0.19*	0.001	0.002/0.38***	0.000	0.000/0.02	0.001	-0.001/-0.20*	0.001	0.002/0.09	0.001	0.001/0.14	0.001
R^2	0.04		0.15***		0.07		0.04		0.01		0.02	
Step 2												
Gender	0.01/0.03	0.01	0.01/0.03	0.02	0.01/0.03	0.02	0.01/0.03	0.02	0.000/0.000	0.02	0.002/0.01	0.02
Culture	-0.001/-0.19*	0.02	0.002/0.38***	0.001	0.000/0.02	0.001	-0.001/-0.20*	0.001	0.000/0.09	0.001	0.001/0.14	0.001
Gender x Culture	-0.001/-0.04	0.001	-0.000/-.005	0.001	0.000/0.02	0.001	0.000/-0.01	0.001	-0.002/-0.16	0.001	0.002/0.14	0.001
ΔR^2	0.002		0.003		< 0.001		< 0.001		< 0.001		0.02	
	RIASEC Model Fit											
	PDI		IDV		PDI		UAI		LTO		IND	
	b/β	*SE*	*b*/β	*SE*	*b*/β	*SE*	*b*/β	*SE*	*b*/β	*SE*	*b*/β	*SE*
Step 1												
Gender[a]	-0.03/-0.09	0.03	-0.03/-0.09	0.02	-0.03/-0.09	0.03	-0.03/-0.09	0.03	-0.03/-0.09	0.03	-0.03/-0.09	0.03
Culture[b]	0.000/-0.05	0.001	0.002/0.26**	0.000	0.001/0.13	0.001	-0.002/-0.24*	0.001	0.000/0.01	0.001	0.001/0.12	0.001
R^2	0.01		0.07		0.02		0.06*		0.01		0.02	
Step 2												
Gender	-0.03/-0.09	0.03	-0.03/-0.09	0.02	-0.03/-0.09	0.03	-0.03/-0.09	0.03	-0.03/-0.10	0.03	-0.03/-0.09	0.03
Culture	0.000/-0.05	0.001	0.002/0.26**	0.001	0.000/0.13	0.001	-0.002/-0.24*	0.001	0.000/0.01	0.001	0.001/0.12	0.001
Gender x Culture	0.000/-0.10	0.001	0.000/-.04	0.001	0.000/0.02	0.002	0.001/0.11	0.001	-0.002/-0.16	0.001	0.000/0.002	0.001
ΔR^2	0.000		0.002		0.000		0.01		0.02		0.000	

n = 59 countries

R^2 for the model in Step 1 represents the total variance explained; ΔR^2 for the model in Step 2 represents the change in variance explained in the model in Step 2 after adding the interaction term

*$p < .05$; ** $p < .01$; ***$p < .001$

[a]gender was effect coded (0.5 = men, -0.5 = women)

[b]Cultural dimensions include: *PDI* power distance, *IDV* individualism (versus collectivism), *MAS* masculinity (versus femininity), *UAI* uncertainty avoidance, *LTO* Long term (versus short term) orientation, *IND* Indulgence (versus restraint)

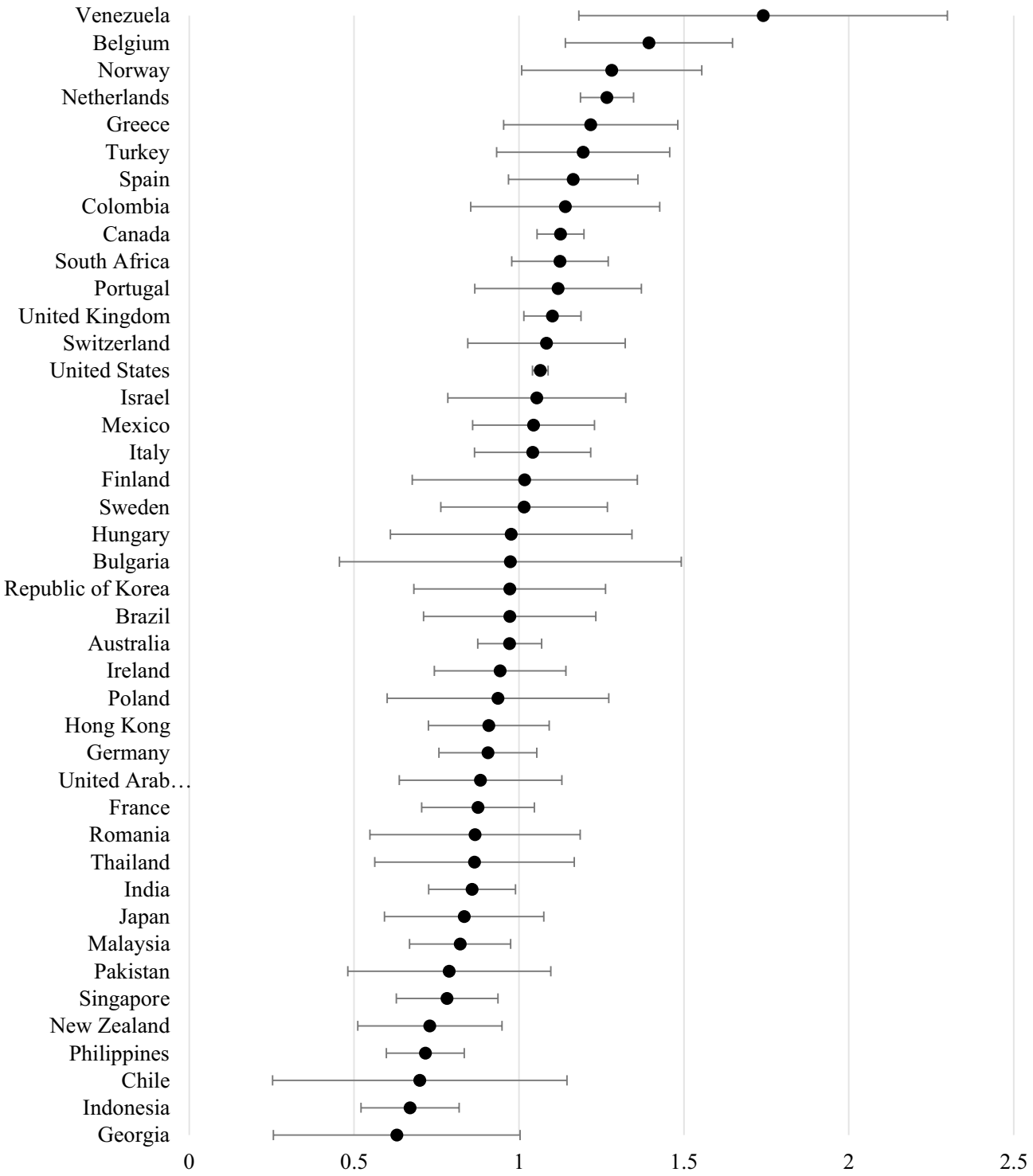


Fig. 3 Forest Plot of Effect Sizes of Gender Differences in Interests in People/Things with 95% Confidence Intervals

sizes of gender differences; gender inequality was negatively associated with differences between men and women (b = -.41, β = -.45, SE = .18, p = .02), such that in countries of higher gender inequality, women's stronger preference for working with people versus things compared to men was smaller. After entering the six national cultural dimensions in the second step of the model, we found that the variance explained increased to 76%. Further, gender inequality was

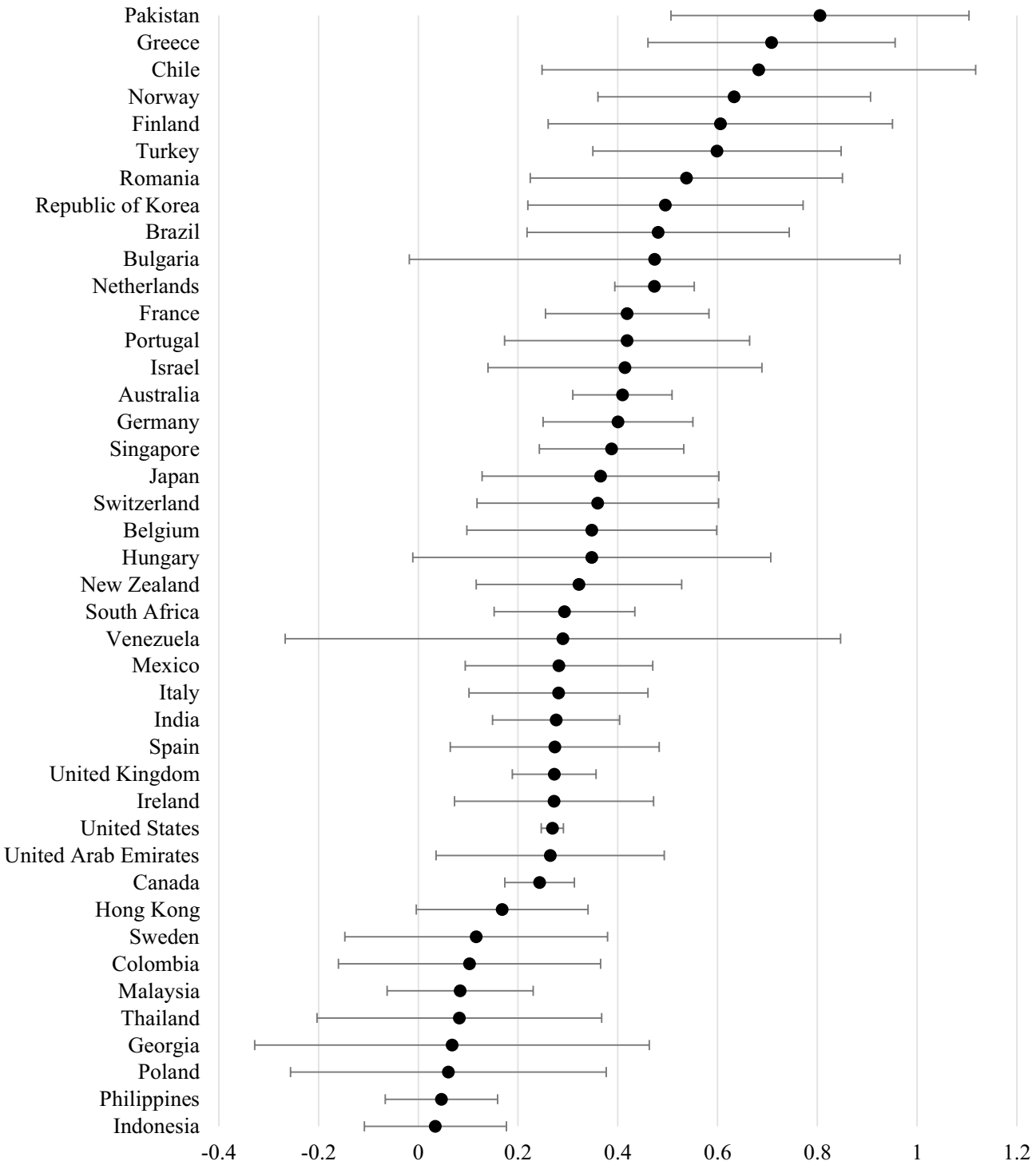


Fig. 4 Forest Plot of Effect Sizes of Gender Differences in Interests in Ideas/Data with 95% Confidence Intervals

not significantly related to the effect size of gender differences in people/things when six national cultural dimensions were considered simultaneously (b = -.06, β = -.07, SE = .17, p = .73). Instead, national uncertainty avoidance was positively related to the effect size (b = .003, β = .47, SE = .001, p = .001), indicating, contrary to our prediction, that those countries with greater uncertainty avoidance had larger differences in interests in people/things between men and women. In addition, national indulgence was positively related (b = .003, β = .55, SE = .001, p = .01) to gender

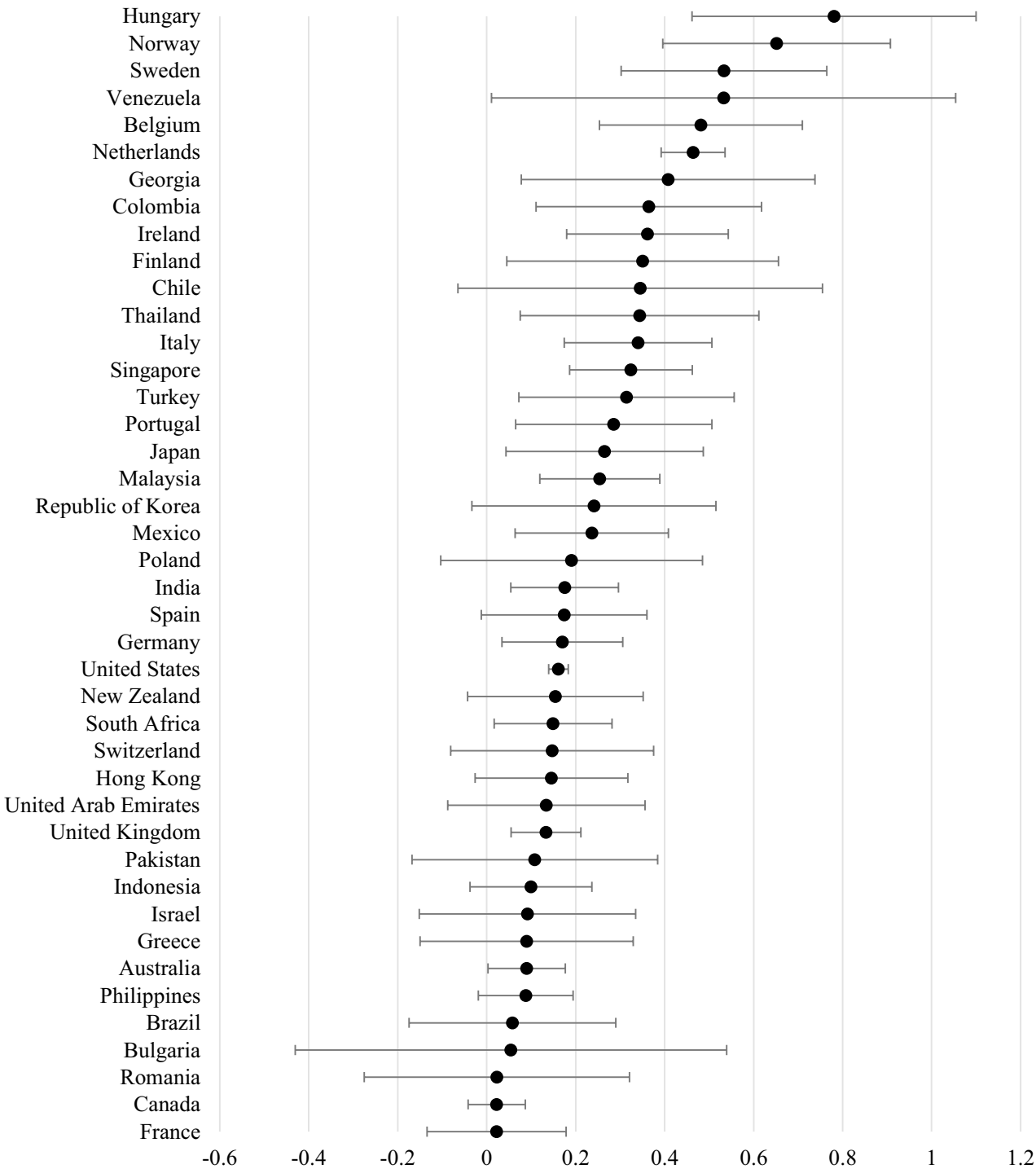


Fig. 5 Forest Plot of Effect Sizes of Gender Differences in Interests in Prestige with 95% Confidence Intervals

differences in people/things, such that countries high on indulgence tended to have larger gender differences in interests in people/things.

For interests in ideas/data, national gender inequality explained 23% of heterogeneity across countries in effect sizes of gender differences, although it was not a statistically significant predictor (b = -.30, β = -.37, SE = .21, p = .16). The addition of the six cultural terms increased the amount of variance explained to a total of 71% of heterogeneity across countries in the effect sizes for gender differences.

Table 3 Meta-analytic Regression Models

	People/Things				Ideas/Data				Prestige			
	b	β	*SE*	*p*	*b*	β	*SE*	*p*	*b*	β	*SE*	*p*
Step 1												
Gender inequality	-0.41	-0.45	0.18	.02	-0.30	-0.37	0.21	.16	-0.22	-0.28	0.13	.10
R^2		0.24				0.23				0.09		
Step 2												
Gender inequality	-0.06	-0.07	0.17	.73	-0.03	-0.04	0.23	.90	-0.09	-0.12	0.14	.55
Power distance	-0.001	-0.14	0.002	.57	-0.004	-0.69	0.002	.01	-0.001	-0.11	0.001	.69
Individualism	0.002	0.33	0.001	.20	-0.001	-0.18	0.001	.52	-0.001	-0.14	0.001	.66
Masculinity	-0.002	-0.31	0.001	.06	-0.002	-0.30	0.001	.07	-0.002	-0.40	0.002	.13
Uncertainty avoidance	0.003	0.47	0.001	.001	0.003	0.50	0.001	.02	0.000	0.05	0.001	.76
Long-term orientation	0.001	0.09	0.001	.65	0.001	0.15	0.001	.43	0.001	0.22	0.001	.32
Indulgence	0.003	0.55	0.001	.01	-0.001	-0.11	0.001	.65	0.002	0.30	0.001	.20
R^2		0.76				0.71				0.36		

Furthermore, when national gender inequality and cultural dimensions were accounted for together, gender inequality was not a statistically significant predictor, (b = -.03, β = -.04, SE = .23, p = .90). In line with our predictions, national power distance had a negative relation with the ideas/data effect size (b = -.004, β = -.69, SE = .002, p = .01), indicating that greater power distance was linked to a smaller gender difference on interests in ideas/data. There also was, again unexpectedly, a positive relation between national uncertainty avoidance and the ideas/data effect size (b = .003, β = .50, SE = .001, p = .02), with greater uncertainty avoidance associated with greater gender differences.

For interests in prestige, national gender inequality explained 9% of variance and the addition of national cultural dimensions explained a total of 36% of variance in the effect sizes for gender differences. However, no statistically significant associations between any of the cultural dimensions and effect sizes for gender differences in interests in prestige were found.

It should be noted that when multiple variables (e.g., national gender inequality, cultural dimensions) were entered in the same model, multicollinearity may affect the findings due to strong correlations between certain variables. However, we had conducted separate analyses with national gender inequality both entered and removed in the model with six cultural dimensions, and there were no notable differences in the direction and statistical significance of the terms in comparison to the models reported.

Discussion

The present study is to our knowledge one of the broadest international studies to date that examines the structure and mean scores of vocational interests by gender. Beyond providing overall international estimates of gender differences of interests in working with people/things, data/ideas, and prestige, we examined the interplay between gender differences between men and women and sociocultural factors, specifically a comprehensive array of national cultural dimensions over and above the frequently examined effects of national gender inequality. Findings also provided broad support for the generalizability of the RIASEC and the octant models to men and women in 42 nations/territories/regions.

Gender Differences in Vocational Interests

With a winnowed dataset of responses from 42 nations demonstrating adequate model fit and after correcting for sample and measurement error, we found a large gender difference in interests in people/things regardless of national context, which is similar to the magnitude of prior findings based on intra-national samples. Consistent with prior research (see e.g., Lippa, 1998; Su et al., 2009; Su & Rounds, 2015; Tracey & Caulum, 2015), women showed a stronger preference of working with people rather than things whereas men preferred working with things rather than people. This finding provides support for widespread if not universal cross-cultural mechanisms contributing to gender differences, as conceptualized by social role and role congruity theories (Eagly & Karau, 2002; Eagly & Wood, 2013; Schmidt, 2014), such that men and women show stronger interests in gender-stereotypical dimensions due to socialization of gender-specific roles and expectations. Similar to previous findings for the gender difference in interest in people/things as one of the strongest differences observed between men and women (Lippa, 1998; Lubinski, 2000; Su et al., 2009), our results based on 42 nations indicate that it might be one of the most culturally universal as well.

Women expressed significantly greater interests in ideas compared to data than men, though this finding is small in

effect size. Although prior findings on gender differences in ideas/data have been mixed (Su et al., 2009), our results appear similar to Morris's finding, based on data using a U.S. based sample, of a small preference for women to work with ideas over data ($d = 0.18$; Morris, 2016) and suggest that such gender differences in ideas/data may exist across national contexts (evidenced in all but two countries in this study). Even though this finding is of a smaller magnitude than gender differences in people/things, this finding is noteworthy and warrants future investigations into factors that may help us better understand men's preference for working with data and women's preference for working with ideas. Although speculative, it is possible that the increasing tendency of digital information, a prominent form of data, to be associated with complex tools (e.g., computers and information systems) orients an interest in data toward an interest in things. Thus, our results on the interest dimension of ideas/data might be a by-product of preferences for people/things. Notably, our findings stand in contrast in certain ways to the earlier findings of Gottfredson (1981) who found that conventional interests, often associated with detail and administrative work (e.g., that of a secretary), were stereotypical for women. Given the shifts in the world of work, the expectations of what would be stereotypical for men and women may change over time, and as such, we may observe different patterns of gender differences in interests as men and women might have been treated differently based on their gender. Future research should explore the implications of the changing nature of work for an understanding of gender stereotypes relating to interests in key work activities.

Gender differences in interest in prestige were also observed, as reflected in men reporting less interest in prestige than women in 34 out of 42 countries included in this study. This finding, albeit a small effect, is noteworthy. It is possible that interests in prestige may be related to individuals' vocational goals (Yalcinkaya & Adams, 2020), such that women may be more interested in pursuing prestige because of their desire to pursue greater financial security associated with a higher status, or to achieve a greater work-family balance or higher level of social impact that prestigious work activities might facilitate. This latter possibility would fit with women's overall greater interest in working with people compared to men (Yang & Barth, 2015).

Taken together, the significant finding and large effect size associated with the gender difference in interests in people/things, in comparison to the effect sizes of interests in ideas/data (3.58 times) and prestige (5.78 times), extend the current largely intra-national findings on gender differences to a much broader international context across 42 nations/ territories/regions. The generalizability of the trends in observed gender differences in interests is noteworthy, though we also observe notable variability in the magnitudes of these differences, which suggests that sociocultural factors may shape individuals' vocational interests.

The Interplay among Cultural Dimensions, Gender, and Vocational Interests

We examined the interactions between national cultural dimensions and individuals' gender on vocational interests beyond the influences of national gender inequality. Existing international research has predominantly focused on how economic development and associated levels of gender equality/egalitarianism might affect gendered differences in vocational interests (Lippa, 2010; Ott-Holland et al., 2013; Stoet & Geary, 2018). This study supplements the literature by investigating six national cultural dimensions beyond these economic and sociocultural factors. Building upon Yalcinkaya and Adams' (2020) cultural-psychological model, we hypothesized that lower power distance, greater individualism, greater masculinity, and lower uncertainty avoidance at the national level would be associated with broadened gender differences in interests after accounting for gender inequality. In the present study, none of the associations between gender inequality and the three interest dimensions was significant after accounting for the six cultural dimensions. Our results thus highlight the importance of assessing culture and its influence on gendered preferences in interests. Furthermore, among the six cultural dimensions examined, we found that national power distance, uncertainty avoidance, and indulgence moderated gender differences in interests. Notably, moderation involving the frequently studied (see e.g., Taras et al., 2010) cultural value of individualism was not detected – yet the less frequently considered cultural dimension of uncertainty avoidance was a moderator of both gender differences in people/things and ideas/data.

Prior research (Taras et al., 2010) has pointed out that national uncertainty avoidance is associated with greater individual neuroticism, commitment to teamwork at work, and preference for direct leadership, but lower levels of innovation; yet few studies have explored its implications for vocational interests. Our hypothesis relating to uncertainty avoidance was based on the reasoning of the cultural psychological model (Yalcinkaya & Adams, 2020), whereby we would have expected national uncertainty avoidance to be inversely associated with gender differences because lower uncertainty avoidance would have been associated with less need to follow rules and greater opportunities for self-expression of interests independent of gender norms, which would reveal a larger gender difference in expression of interests. However, our results illustrated the opposite pattern. Those who live in more clearly rule-oriented societies (i.e., higher national uncertainty avoidance) may be more inclined to conform to gender stereotyped vocational

interests (e.g., women more likely than men to favor working with people versus things and ideas versus data). In other words, in nations where certainty is highly valued, the social structure may influence residents' gender roles to a greater extent so that they have less agency and/or have a lesser desire to deviate from stereotypical gendered patterns of interests. Indeed, existing research has shown similar findings at the individual level such that those who are more highly avoidant of uncertainty tend to hold more conservative and traditional views and values (see e.g., Jost et al., 2007). Therefore, this result extends the conceptualization to societal/national influences of uncertainty avoidance as individuals in more ambiguity avoidant countries appear to follow more gender stereotypical roles (Hofstede et al., 2010). Further, the result provides support for the social role hypothesis (Falk & Hermle, 2018), which emphasizes the extent to which gender-role expectations shape preferences and behavior. We hasten to add that this unexpected finding could be due to the difference in measurement of interests or cultural dimensions and the control of effects of national gender inequality and other cultural dimensions. Nevertheless, it suggests that the logic of the resource hypothesis may not be relevant to all cultural dimensions with varying patterns of gendered differences in interests.

Support for the resource hypothesis (Falk & Hermle, 2018) in relation to power distance was found, but support was limited. Although power distance does not necessarily imply gender inequality, it reflects the extent to which unequal power is generally accepted at the societal level. Moreover, societal power (e.g., political and economic power) is unequally distributed between men and women in every country (United Nations Development Programme, 2020). Thus, in line with the resource hypothesis, our results indicate that when accounting for other cultural dimensions, men and women tend to express less pronounced gender differences in interests in ideas and data when power distance is higher. Few studies to date have explored the mechanisms between broad social equality and vocational outcomes. Two studies are an exception to this (Costa et al., 2001; Guimond et al., 2007). In these studies, it was shown that the more men and women compare themselves to each other in certain cultural contexts, the more they may engage in self-stereotyping which may be associated with greater gender differences (Costa et al., 2001; Guimond et al., 2007). Building upon the insights of these studies, it is possible that gender differences in interests within societies of higher power distance are smaller between men and women due to role expectations shaped by cross-gender comparisons. Future research should examine this possibility.

Explorations on the effect of national indulgence reveal that the gender difference in interests in people/things is more prominent in countries higher in national indulgence (versus restraint). This finding may be explained by the cultural psychological model (Yalcinkaya & Adams, 2020). Greater indulgence and less restraint at a national level may have promoted greater freedom of choice and self-expression of individuals, which may in turn have propelled men and women to pursue interests more congruent with their gender roles that result in a greater identity fit. In other words, men and women might be more likely to indulge career desires that arise from their gendered identities and less likely to be influenced by other factors that may restrain their interests (e.g., a need to fulfill financial security).

In summary, although prior findings on gendered preferences or outcomes have largely supported the resource hypothesis (Falk & Hermle, 2018), our findings using a broader set of cultural values provided only mixed evidence for related reasoning from the cultural psychological model (Yalcinkaya & Adams, 2020) – with some support provided instead for the social role hypothesis (Falk & Hermle, 2018). Nevertheless, our results reinforce the potential that different cultural values beyond individualism affect vocational interests between men and women in unique ways. Moreover, our results provide unique evidence that not one single theory or hypothesis may explain how certain cultural values affect individuals' interests (Charles & Bradley, 2009; Stefani & Prati, 2021). Accordingly, sociocultural environments at the national level may be associated with men and women's vocational interests both directly and indirectly. Therefore, it is important to recognize the potential for national cultural dimensions to affect vocational interests in ways beyond the effects of the more frequently examined factors of modernization, economic development, and national gender (in) equality.

Limitations

Findings from this study should be interpreted considering several limitations. First, although our findings are based on a large dataset of responses in an international context, the responses were obtained from a self-selected group that had access to *Time* magazine's website. Moreover, there is substantial variability in sample sizes across countries, varying from a little over 60 responses from Qatar to over 40,000 responses from the United States. Thus, responses in each country may not be representative of all individuals in each country, especially in countries where English was not the native language and where not all individuals have access to a computer, a smart phone, or the Internet. Despite this limitation, we accounted for variability in responses across countries in our meta-analytical regression analysis to minimize the impact of sampling bias. Future studies can benefit from more targeted recruitment of respondents.

A second limitation of our study was that the determination of nationality was based on the IP address of responses. This was a practical approach in a large-scale international

study; however, we had no indication of how long individuals had lived in the respective nation. It is thus possible that our dataset included, for example, some individuals who were not long-term residents of these countries. However, these individuals were likely residing at the time in their assigned country. Rentfrow et al. (2008) and Schmidt et al. (2007) have pointed out that individuals can show differences in psychological characteristics by geography. Although such differences can be due to inherent differences in local residents, they can also be due to individuals being drawn to certain regions because of similar characteristics to local residents. As such, the possibility of some respondents not being citizens of a country does not necessarily invalidate their responses.

Responses in the study are tied to individuals who are relatively highly educated and appear to be better off financially than the average in most countries. Thus, our results may apply only to the members of higher socioeconomic status in each country. Furthermore, having notable missing data on respondents' annual income (23.30%), and relying upon an operationalization of income solely based on dollar values also limits our understanding of how responses in our study compare to the average annual income of respondents' respective country of residence. Certainly, more work is needed to recruit broader and more diverse samples.

Additionally, individuals responded to our demographic survey by selecting either "man" or "woman," and it is unclear if transgender or gender non-binary persons participated in this study. Emerging research suggests that transgender individuals may face unique challenges with their vocational decisions and self-efficacy due to factors like gender identity, transition status, and experiences of discrimination (see e.g., Dickey et al., 2016; Schneider & Dimito, 2010). Prospective studies can broaden the examination of gender differences by including samples of self-identified transmen and transwomen and gender non-binary individuals to explore the structure of their vocational interests as well as the sociocultural influences.

Finally, given the cross-sectional nature of this examination, we cannot infer the directional influences regarding the associations among national cultural dimensions, gender, and vocational interests, or on interest stability over time. Such associations can only be unraveled with a longitudinal design. Indeed, it is possible that these associations may differ depending on a cross-sectional or longitudinal design of research (Connolly et al., 2020).

Future Directions

Our examination highlights broader sociocultural influences that should be considered in gender studies, especially as they affect vocational interests in unique ways when gender (in)equality at the national level is considered. Hofstede's cultural value framework (Hofstede, 1984, 2016; Minkov & Hofstede, 2012b) is a well validated representation of dimensions on which cultures vary. However, the scores of cultural dimensions for each country represent the mean values of individuals in each country and only serve to characterize between-country differences. That is, national dimensional indices may not reflect the extent to which individuals internalize or believe these values in similar ways as the broader society that they live in. Future studies could examine not only individual perceptions of national values but also personal values. That way these cultural variables can completely represent the extent of within and between differences. Potential interactions between individual and societal values can also be explored.

Another promising direction for future research is to better understand if and how national cultural values are embraced by individuals themselves and influence their daily experiences, which can reflect the acculturation and enculturation process (Kantamneni & Fouad, 2013). Such insights can relate to individuals' educational and career choices. There is a need to better understand individuals' challenges and strengths in the face of educational and career exploration from a culturally informed perspective. Doing this can help to reveal potential intra-national and interpersonal variations in the associations between vocational interests and cultural dimensions at the societal level. Emerging evidence suggests that individuals' perceived similarity and belonging with others in certain careers are linked to interests (Cheryan & Plaut, 2010; Tellhed et al., 2017), which may in turn, be linked to educational and career decisions and performance (see e.g., Rounds & Su, 2014; Tracey & Robbins, 2005; Tracey et al., 2005).

Another important consideration is that sociocultural factors may affect the influence of vocational interests on career choices. Given the focus on gender differences in vocational interests in this study, less is known about how these gender differences may influence individuals' career-related decisions and how culture may affect this mechanism. For example, individuals in more collectivistic cultures may place a greater emphasis on financial security and needs for the individual and their family beyond interests (Charles & Bradley, 2009; Yalcinkaya & Adams, 2020). Therefore, researchers are encouraged to explore additional socio-environmental factors related to work (e.g., a sense of belonging) in addition to national cultural dimensions to explore the impact of culture on interests, and educational and career outcomes by gender.

Practice Implications

Beyond research implications, findings from this study can offer important practical implications for vocational psychologists, school counselors, and career counselors. These

professionals may benefit from the consideration of both gender and cultural effects when working with individuals to explore educational and career options and experiences. Specifically, professionals can consider using interest and career assessment tools. Measures like the Personal Globe Inventory-Mini that have been deemed appropriate for use across a broad array of countries can be used with individuals who grew up or live in diverse nations/territories/regions. Equally important is to conceptualize results from an interest assessment with a consideration of cultural influences. Professionals can guide individuals through in-depth conversations about career exploration in the context of individuals' gender identity and cultural environments (Flum & Blustein, 2000). When assessing the nature and implications of individuals' vocational interests, psychologists and counselors are encouraged to not only refer to the national cultural dimensions as evaluated by researchers like Hofstede and colleagues (Hofstede, 1984; Minkov & Hofstede, 2012a, 2014), but inquire about individuals' personal perceptions and associated narratives. These discussions can involve multiple dimensions, such as how uncertainty in societal structure is treated, how power is distributed amongst people of diverse backgrounds, how masculinity and femininity is viewed, the extent to which individualism and collectivism are valued, and the extent to which human drives are valued to be indulged or restrained.

Conclusion

In conclusion, the present study supports the overall generalizability of the RIASEC and octant models for men and women in 42 nations/ territories/regions. Compared to men, women endorse a general trend of a large preference for working with people (versus things), and smaller preferences for working with ideas (versus data) and with prestige. Our findings lend tentative support for the effects of national cultural dimension on gender differences in vocational interests beyond the effects of national gender inequality. In fact, the interplay between gender and diverse national cultural dimensions appears to be nuanced and partially supportive of the social role hypothesis (Falk & Hermle, 2018), the cultural psychological model (Yalcinkaya & Adams, 2020), and the resource hypothesis (Falk & Hermle, 2018). These findings call for advancements in our theoretical frameworks to better characterize the cultural influences on vocational interests beyond the predominant focus on economic modernization, gender equality, or single cultural dimensions. Together, our findings encourage prospective research and professional practices to consider the unique roles that specific national cultural values, such as uncertainty avoidance, may play in the vocational interests of men and women.

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Compliance with Ethical Standards

Conflict of Interest We have no known conlfict of interest to disclose. However, we want to make note that the Personal Globe Inventory-Mini (PGI-Mini) used in this study was developed by the third author, Terence J. G. Tracey, and has copyrighted. It is available for free at http://doi.org/10.23668/psycharchives.4545.

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